Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Ryan, Margaret Amy Ph.D.		
Project Title:	Second Generation Electronic Nose		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	818-354-8028
Organization Name:	NASA Jet Propulsion Laboratory		
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PI Web Page:	http://enose.jpl.nasa.gov		
City:	Pasadena	**State:**	CA
Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	10/01/1998	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
	The Jet Propulsion Laboratory (JPL) Electronic Nose (ENose) program is developing a broad band air quality monitor which will facilitate crew health and safety in the Space Station. Successful completion will result in an incident monitor capable of real time air quality monitoring. This device, or a group of such devices deployed as distributed sensors, will monitor changes in air composition and return information on the identity and quantity of compounds causing the change. The monitor will determine whether the change involves contaminants on the target list, and the quantity of contaminant in concentrations related to Spacecraft Maximum Allowable Concentrations (SMACs). The JPL ENose is a low mass, low power device which requires little crew time for maintenance and calibration. JPL's ENose is a miniature environmental monitoring instrument that detects and identifies a wide range of organic and inorganic molecules down to the parts-per-million level. The kinds of changes that the ENose is designed to detect are anomalous events such as leaks and spills. The monitoring device uses an array of electronically conducting, thin-film polymer-carbon composite sensors. The chemical sensors change resistance when exposed to vapors by swelling or		

Task Description:	shrinking when vapor molecules enter or leave the film. No one sensor is dedicated to identifying any one compound. The pattern developed by the change in resistance of the entire array of sensors is used to identify the compound, and the magnitude of the response pattern is used to quantify the compound. Deviations from a baseline of clean air, determined by recording the response of the array to air cleaned in an activated carbon filter, are identified and quantified using a software program designed for this use. The need for a low power, autonomous air quality monitor is pressing for an enclosed space such as the crew quarters of the International Space Station (ISS). In such a space, the presence of contaminants is particularly threatening because the crew is entirely reliant on the air revitalization systems (ARS) for healthy air; the air is not easily replaced. Contaminants may enter the air through leaks and spills from equipment and experiments or from smoldering fires. This kind of incident is a threat to crew health because they are unexpected and can occur with a sudden, large increase in contaminant concentration that may be toxic. The earliest warning possible is necessary to avoid threat to crew health from such incidents. Ideally, monitors which provide early warning of danger to crew health in the breathing air would be interfaced to ISS controls for automatic cleanup, sealing off affected areas until the crew can clean it up, or alerting the crew. The goal of this technology development program is development of enabling technology for an integrated environmental monitoring and control system for the International Space Station and beyond. This program addresses the first step toward that goal, the development of an Autonomous Electronic Nose. Phase I has been completed and resulted in a prototypic model ENose with the capability to detect 10 compounds at the 1 hour SMAC. This device was flown as an experiment on STS-95 in 1998. We are now in Phase II of this development, and will move to Phase III in the coming fiscal year. The technical objective of Phase II is the completion of all technology components necessary for an Autonomous ENose, including optimized sensors, data acquisition hardware and software, and data analysis hardware and software, packaged suitably for flight or ground experiments. Phase III is extensive ground testing in conditions which will challenge all the aspects of the ENose: identification and quantification of 20-25 compounds at the 24 hour SMAC, ability to reject unknowns, and deconvolution of mixtures of up to 4 target compounds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ability to monitor the air quality in an enclosed space has numerous applications on Earth. A reliable electronic nose will be beneficial to human safety in enclosed spaces such as aircraft and submarines. In addition, the ability to monitor air for contaminants such as coolant leaks and chemical process leaks in manufacturing environments will be beneficial to human and instrument safety. Further development of the device may make it possible to use an electronic nose in such applications as pollution monitoring, explosives detection, and in medical diagnosis.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Articles in Peer-reviewed Journals	Ryan MA, Zhou H, Buehler MG, Manatt KS, Mowrey VS, Jackson SP, Kisor AK, Shevade AV, Homer ML. "Monitoring space shuttle air quality using the Jet Propulsion Laboratory electronic nose." IEEE Sens J. 2004 Jun;4(3):337-47. PubMed PMID: 15875356 , Jun-2014